

DPA System – End-to-End Project Plan (Claude Execution Plan)

0. Purpose

This document provides a **step-by-step execution plan** for building the Data Protection Automation System (DPA System).

Claude must follow this plan sequentially and produce deliverables at each stage.

1. Execution Principles

- Build iteratively (MVP → production)
 - Every phase must produce **usable output**
 - Prefer **working code over theory**
 - Keep components modular and API-driven
 - All decisions must be **traceable and auditable**
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2. Phase Overview

Phase	Name	Outcome
1	System Architecture	Full system blueprint
2	Data Modeling	Database schema
3	Rule Engine	Decision logic framework
4	Workflow Engine	Orchestration layer
5	Backend APIs	Functional backend
6	Frontend UI	User interface
7	Integrations	External system connectivity
8	Compliance Logic	Regulatory enforcement
9	Audit & Logging	Traceability
10	Deployment	Production-ready system

3. Phase 1 – System Architecture

Objective

Design a complete system architecture.

Tasks

- Define system components
- Define service boundaries (microservices vs monolith)
- Define data flow between components
- Define deployment model (cloud, containers)

Deliverables

- Architecture diagram (text-based)
 - Component breakdown
 - Technology stack selection
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4. Phase 2 – Data Modeling

Objective

Design all core data structures.

Tasks

- Identify core entities:
 - Use Cases
 - Assessments (PIA, TIA, DSA)
 - Users
 - Approvals
 - Rules
 - Workflows
 - Audit logs
- Define relationships
- Define retention + metadata fields

Deliverables

- ER diagram (textual)
 - SQL schema (PostgreSQL preferred)
 - Sample records
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5. Phase 3 – Rule Engine

Objective

Build a decision engine that evaluates compliance rules.

Tasks

- Define rule format (JSON or DSL)
- Implement rule evaluator
- Define conditions and outputs

Example Rules

- Personal data + marketing → require consent
- AI + credit scoring → high risk
- Data outside EEA → trigger TIA

Deliverables

- Rule schema
 - Rule engine implementation
 - Sample rules
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6. Phase 4 – Workflow Engine

Objective

Create orchestration for approvals and processes.

Tasks

- Define workflow states
- Define transitions
- Implement approval chains
- Add SLA tracking

Deliverables

- Workflow model
 - State machine logic
 - Example workflows
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7. Phase 5 – Backend APIs

Objective

Expose system functionality via APIs.

Tasks

- Design REST or GraphQL APIs
- Implement endpoints:
- Create assessment
- Evaluate rules
- Trigger workflows
- Approvals
- Audit logs

Deliverables

- API specification
 - Backend implementation (Node.js or Python)
 - Example requests/responses
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8. Phase 6 – Frontend UI

Objective

Build user interface for assessments and monitoring.

Tasks

- Create dynamic forms (based on questionnaires)
- Build dashboards
- Build approval interfaces

Deliverables

- UI structure
 - Component list
 - Sample screens (React preferred)
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9. Phase 7 – Integrations

Objective

Connect to external systems.

Tasks

- Define integration patterns (API, webhook)
- Simulate connections to:
 - CRM
 - Data platforms
 - Identity systems

Deliverables

- Integration architecture
 - Example connectors
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10. Phase 8 – Compliance Logic

Objective

Encode regulatory frameworks.

Tasks

- Implement:
 - Privacy (GDPR)
 - DSA
 - AI risk classification
 - Data transfer safeguards

Deliverables

- Compliance rule sets
 - Risk scoring model
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11. Phase 9 – Audit & Logging

Objective

Ensure full traceability.

Tasks

- Log every action
- Store decision reasoning
- Enable audit queries

Deliverables

- Audit schema
 - Logging implementation
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12. Phase 10 – Deployment

Objective

Prepare system for production.

Tasks

- Containerize services (Docker)
- Define CI/CD pipeline
- Configure environment variables

Deliverables

- Deployment architecture
 - Docker configs
 - Run instructions
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13. MVP Scope (Important)

Claude must first build an MVP including:

- Basic use case intake form
 - Rule engine with 5–10 rules
 - Simple workflow (approve/reject)
 - API backend
 - Minimal UI
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14. Execution Instructions for Claude

Claude should: 1. Complete one phase at a time 2. Output all deliverables before moving on 3. Provide working code where applicable 4. Avoid placeholders unless necessary 5. Ensure each phase builds on the previous one

15. Final Output Expectation

By the end, Claude must produce:

- A working MVP system
 - Clear architecture
 - Extendable compliance framework
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FINAL COMMAND

Follow this project plan step-by-step and begin with **Phase 1: System Architecture**.